FDS EXPERIMENT-3

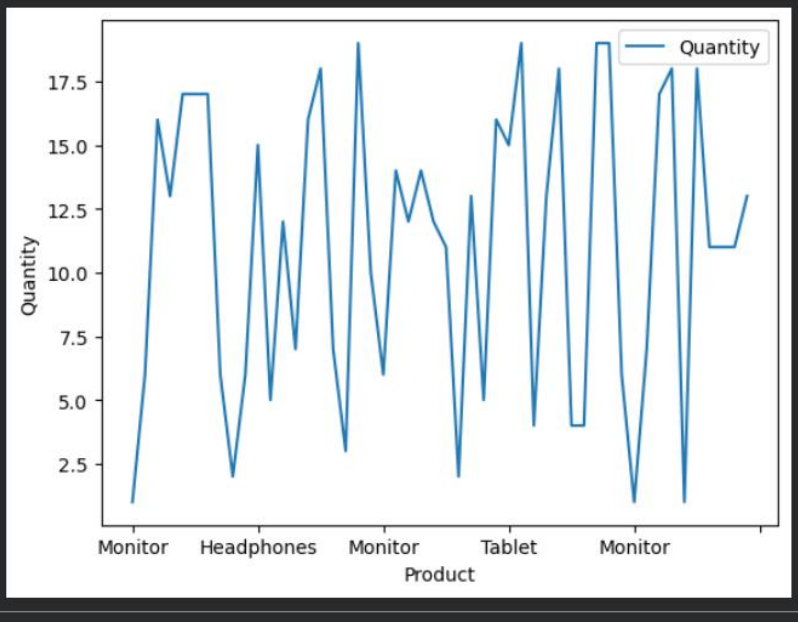
SUBIN P - 250701750

3a) Conduct an experiment to show data visualization using line plot.

```
import pandas as pd
df = pd.read_csv("/content/clean_sales_dataset (1).csv")
print(df)
```

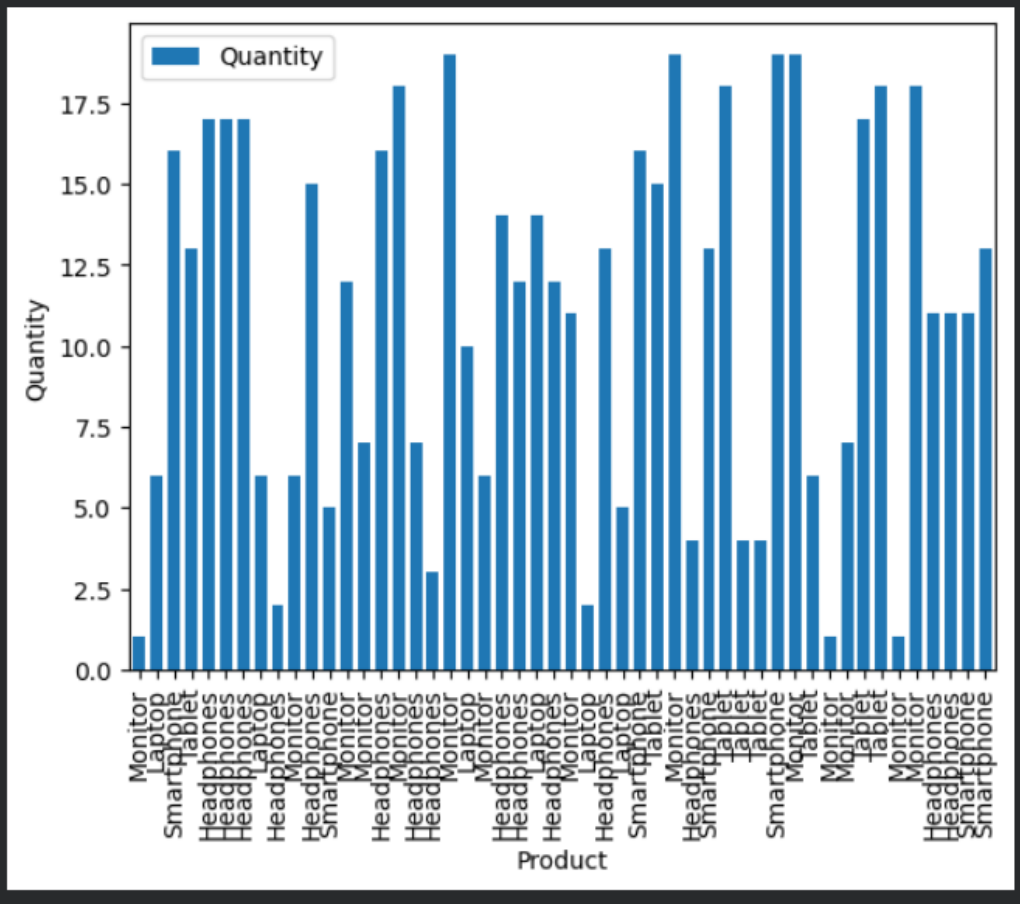
```
    Serial no        Date     Product     Sales  Quantity Region or place
0           1  2023-12-15     Monitor    261.18         1            West
1           2  2023-12-24      Laptop   5989.55         6            West
2           3  2023-07-20  Smartphone   9576.29        16           South
3           4  2023-08-14      Tablet   5248.49        13            East
4           5  2023-08-06  Headphones   1670.38        17            East
5           6  2023-01-17  Headphones   1708.08        17            West
6           7  2023-04-28  Headphones   1720.41        17           North
7           8  2023-09-22      Laptop   6003.06         6           North
8           9  2023-05-17  Headphones    195.65         2           North
9          10  2023-06-26     Monitor   1453.36         6           South
10         11  2023-03-11  Headphones   1436.12        15           South
11         12  2023-03-08  Smartphone   3058.45         5           North
12         13  2023-01-12     Monitor   2993.82        12            West
13         14  2023-01-12     Monitor   1741.73         7           North
14         15  2023-05-30  Headphones   1637.97        16           South
15         16  2023-08-17     Monitor   4509.30        18           South
16         17  2023-09-08  Headphones    705.39         7           South
17         18  2023-11-03  Headphones    314.03         3            East
18         19  2023-11-15     Monitor   4770.69        19            East
19         20  2023-11-24      Laptop  10010.62        10           North
20         21  2023-12-25     Monitor   1514.17         6           North
21         22  2023-07-10  Headphones   1389.38        14            East
22         23  2023-04-26  Headphones   1219.86        12           North
23         24  2023-12-25      Laptop  13972.62        14           South
24         25  2023-02-13  Headphones   1226.17        12           North
25         26  2023-02-02     Monitor   2762.75        11            West
26         27  2023-04-16      Laptop   2014.56         2           South
27         28  2023-05-13  Headphones   1348.99        13           South
28         29  2023-01-14      Laptop   5005.74         5            East
29         30  2023-10-27  Smartphone   9588.37        16           South
30         31  2023-03-26      Tablet   6008.84        15           North
31         32  2023-03-14     Monitor   4739.47        19            East
32         33  2023-08-27  Headphones    373.89         4            West
33         34  2023-10-11  Smartphone   7774.64        13            West
34         35  2023-01-24      Tablet   7211.27        18           South
35         36  2023-03-26      Tablet   1583.75         4           South
36         37  2023-05-14      Tablet   1574.01         4           South
37         38  2023-10-27  Smartphone  11410.81        19           South
38         39  2023-12-04     Monitor   4764.33        19            West
39         40  2023-09-10      Tablet   2377.68         6            East
40         41  2023-08-05     Monitor    249.40         1           South
41         42  2023-11-10     Monitor   1758.55         7            East
42         43  2023-06-27      Tablet   6821.06        17            West
43         44  2023-02-13      Tablet   7179.26        18           North
44         45  2023-09-15     Monitor    243.48         1           North
45         46  2023-08-23     Monitor   4565.47        18           South
46         47  2023-07-26  Headphones   1085.14        11           North
47         48  2023-09-17  Headphones   1085.51        11            East
48         49  2023-01-10  Smartphone   6576.87        11           North
49         50  2023-09-08  Smartphone   7803.01        13            East
```

```
import matplotlib.pyplot as plt
df.plot(kind='line',x='Product',y='Quantity')
plt.xlabel('Product')
plt.ylabel('Quantity')
plt.show()
```



3b) Conduct an experiment to show data visualization using bar chart.

```
[ ]
import matplotlib.pyplot as plt
df.plot(kind='bar',x='Product',y='Quantity',width=0.7)
plt.xlabel('Product')
plt.ylabel('Quantity')
plt.show()
```



3c) Conduct an experiment to show data visualization using pie chart.

```
import matplotlib.pyplot as plt
product_quantity = df.groupby('Product')['Quantity'].sum()
fig, ax = plt.subplots(figsize=(8, 8))
ax.pie(product_quantity, labels=product_quantity.index, autopct='%1.1f%%', startangle=90, textprops={'fontsize': 10})
ax.axis('equal')
plt.title('Distribution of Quantity by Product', fontsize=14)
plt.show()
```

